AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 4 · WORKBOOK

Rules or Learned Patterns?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected fitting record.

Execute a rule, compare candidates, inspect a tie, protect the final check and fit a different classifier.

Keep one habit: Keep predictions, labels, error counts and the reason for the chosen rule.

MAIN LESSON · SECTION 1

Recall · Measurements and mechanisms

> Retrieve both earlier ideas before checking your notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternative practice and picture transcripts**

Sensors Are Messengers

AI or Not?

MAIN LESSON · SECTION 12

W1 · Execute the boundary

> Apply the stated comparison before selecting any new rule.

Your evidence record

Use wide when width is at least 5 cm. Predict the **label** for widths 4, 5 and 6. Then state what changes if the rule uses greater than 5 instead.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compare the candidate errors

> Write every prediction before counting disagreements.

Your evidence record

Widths 1, 3, 5, 7 have **label**s narrow, narrow, wide, wide. For candidates 3 and 5, list all **prediction**s using wide when width is at least the candidate. Count disagreements and select the lower-error candidate.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Make a tie visible

> Use the supplied fitting examples and identify a new input that separates the candidates.

Your evidence record

For the worked widths 2, 4, 6, 8 with **label**s narrow, narrow, wide, wide, compare **threshold**s 5 and 6. Give a width where they predict differently. Explain why the four **fitting** examples cannot decide that case.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Keep the check honest

> Explain when a revealed check case becomes development data.

Your evidence record

A learner sees that their final-check card was predicted incorrectly, changes the **threshold** to match it and reports the same card as an unseen test. Explain the problem and propose a valid next check.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Fit a new length classifier

> Keep one first attempt before hints or teacher feedback.

Read and explain independently

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Fit a new paper classifier

Project · Keep the rule's origin

> Record the written design or candidate-error evidence.

Your project decision

- Draw a fixed-rule version and a learned-pattern version of the Classroom Sorter; state what **evidence** would justify the learned version.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep your original attempt and a later changed-case check.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to a prediction list, fitting choice and limit.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported comparison copy**

REFERENCE · SECTION 2

Reference · Supported comparison copy

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Complete the supported comparison for Rules and **learned pattern**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Can a learned rule be simple?

REFERENCE · SECTION 3

Extra practice · Can a learned rule be simple?

> Use a fitted threshold as your evidence.

Use this when helpful

Use the worked case for Rules and **learned pattern**s. Proposed explanation: “A learned rule cannot be written simply”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Same output, different origin

REFERENCE · SECTION 5

Reference · Same output, different origin

> Use the purple-card question as an explanation target.

Use this when helpful

Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

▼ **Extra practice · Missing blue-card rule**

REFERENCE · SECTION 6

Extra practice · Missing blue-card rule

> Do not invent handling that was never specified.

Use this when helpful

Sorter A follows a written red-card-left rule. Sorter B is fitted using **label**led leaves.

How were their decision **process**es obtained? Where should A send blue cards?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Written and fitted rules**

REFERENCE · SECTION 7

Extra practice · Written and fitted rules

> Keep this practice separate from the reserved new classifier.

Use this when helpful

System A uses the **written rule** green means go. System B learns fruit **label**s from example photographs. Who supplies A's rule? What changes when B is trained on different examples?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Purple-card response space**

REFERENCE · SECTION 11

Reference · Purple-card response space

> This teaching case is practice rather than fresh assessment.

Use this when helpful

Sorter A follows 'red left, blue right'. Sorter B was trained using **label**led red and blue cards. Both send a purple card left. Does their shared **output** show that they used the same **process**? What **evidence** is missing?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · New classifier response copy

REFERENCE · SECTION 15

Reference · New classifier response copy

> This duplicates W4. Reuse its first-attempt record.

Use this when helpful

A fictional sorting task has lengths 2, 5, 7, 9 cm with **label**s short, short, long, long. Candidate rules predict long when length is at least 5 or at least 7.

List **prediction**s and errors for both, choose one, and predict a new length of 6 before any **label** is revealed. State one limit of your conclusion.

> **My reasoning and evidence**

▼ Extra practice · Read the pebble trace

REFERENCE · SECTION 18

Extra practice · Read the pebble trace

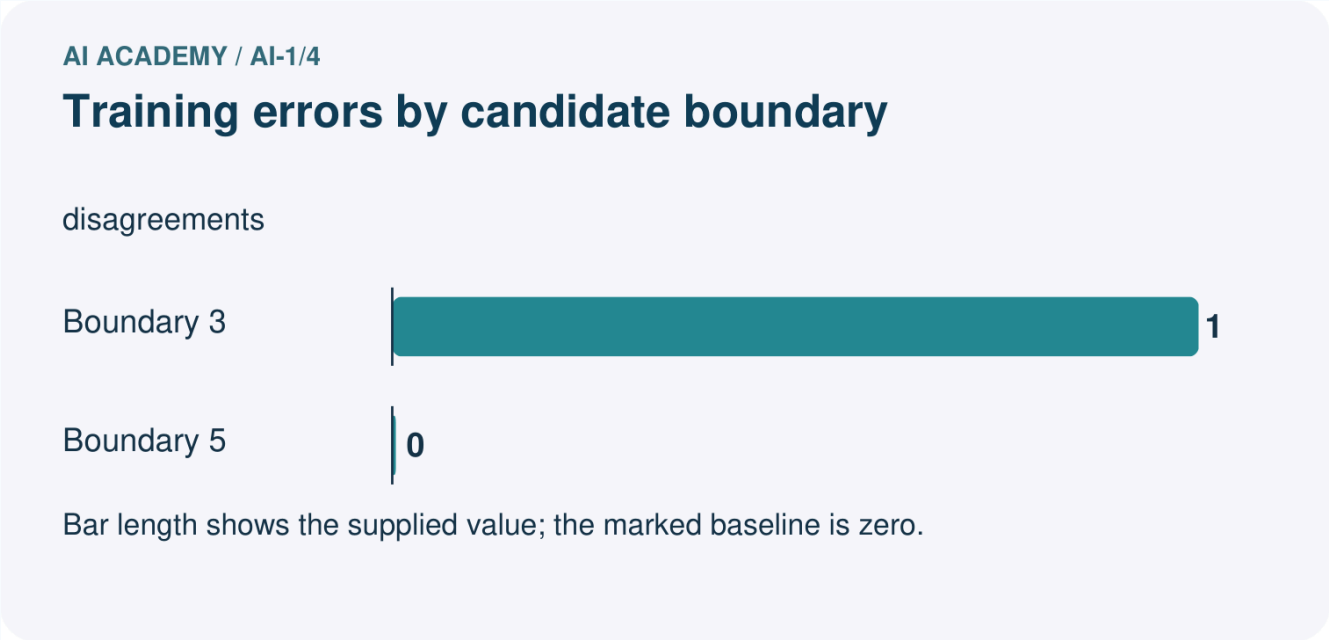
> Use the worked picture and record your support.

Use this when helpful

A fictional paper sorter learns a size boundary for toy pebbles.

Training diameters are 1,3,5,7 cm with **label**s small,small,large,large.

Compare large-at-least-3 with large-at-least-5; choose fewer errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/4

Rules or Learned Patterns?

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Training examples
Diameters 1,3,5,7 have reference labels S,S,L,L.



2. Candidate 3
Predictions S,L,L,L: one disagreement at 3.



3. Candidate 5
Predictions S,S,L,L: zero disagreements.



4. New pebble
Selected boundary 5 predicts small for a new diameter 4.

CHANGE ONE THING
Give a new pebble diameter 5 instead of 4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Show which stage used **label**s and which stage used only the new measurement.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors.

▼ **Reference · Sorter portfolio record**

REFERENCE · SECTION 19

Reference · Sorter portfolio record

> Extend your main project record if needed.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: fitting pictures and feedback

REFERENCE · SECTION 20

Extra practice: fitting pictures and feedback

> Try before opening each explanation. Keep these supported questions separate from the independent case.

First supported question

- A fictional paper sorter learns a size boundary for toy pebbles.
- Which **label**led step matches this detail? Selected boundary 5 predicts small for a new diameter 4.
- 1. **Training** examples
- 2. Candidate 3
- 3. Candidate 5
- 4. New pebble

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: New pebble Selected boundary 5 predicts small for a new diameter 4. The learning procedure compares candidate rules against supplied examples and selects boundary 5.

Applying that boundary to diameter 4 is a later **prediction** step. An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable.

Changed-boundary question

- Changed-case practice: Give a new pebble diameter 5 instead of 4. Which conclusion is supported by the supplied facts?
- 1. The selected rule predicts large.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the boundary**

After your attempt, compare with the supported explanation: The selected rule predicts large. At least includes equality, so 5 meets the boundary.

Repair and save

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternative practice and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

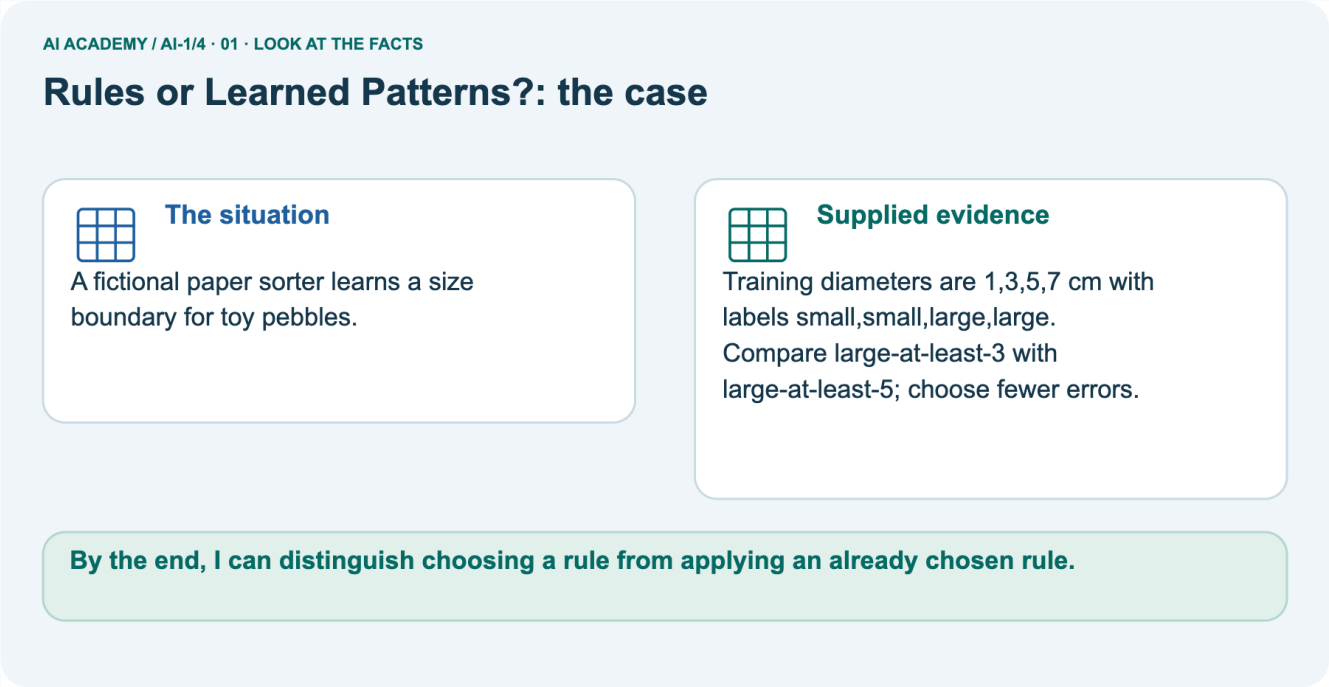
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose a relevant picture rather than repeating all six.

01. Rules or Learned Patterns?: the case



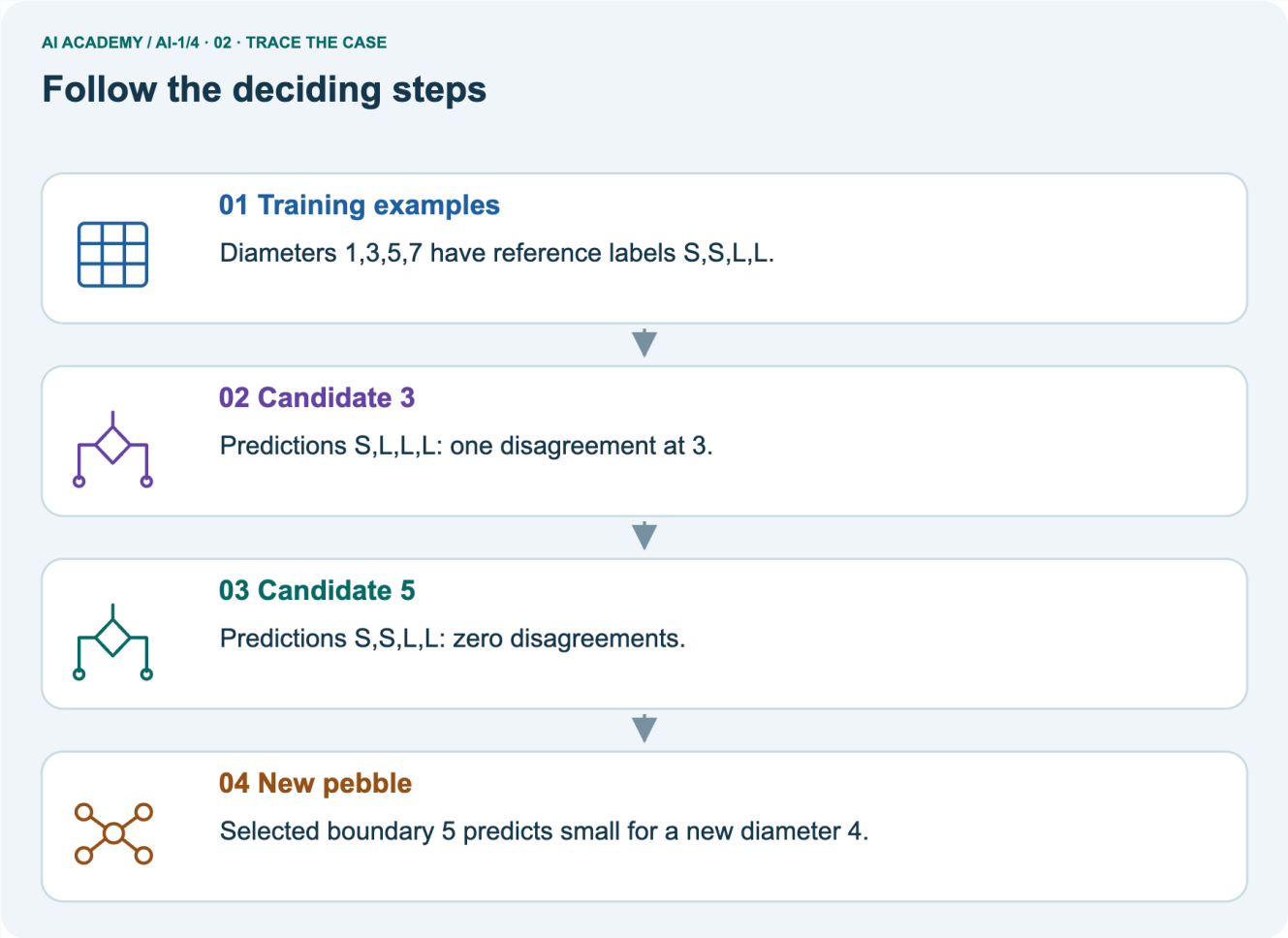
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter learns a size boundary for toy pebbles. Supplied facts: **Training** diameters are 1,3,5,7 cm with **label**s small,small,large,large. Compare large-at-least-3 with large-at-least-5; choose fewer errors.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



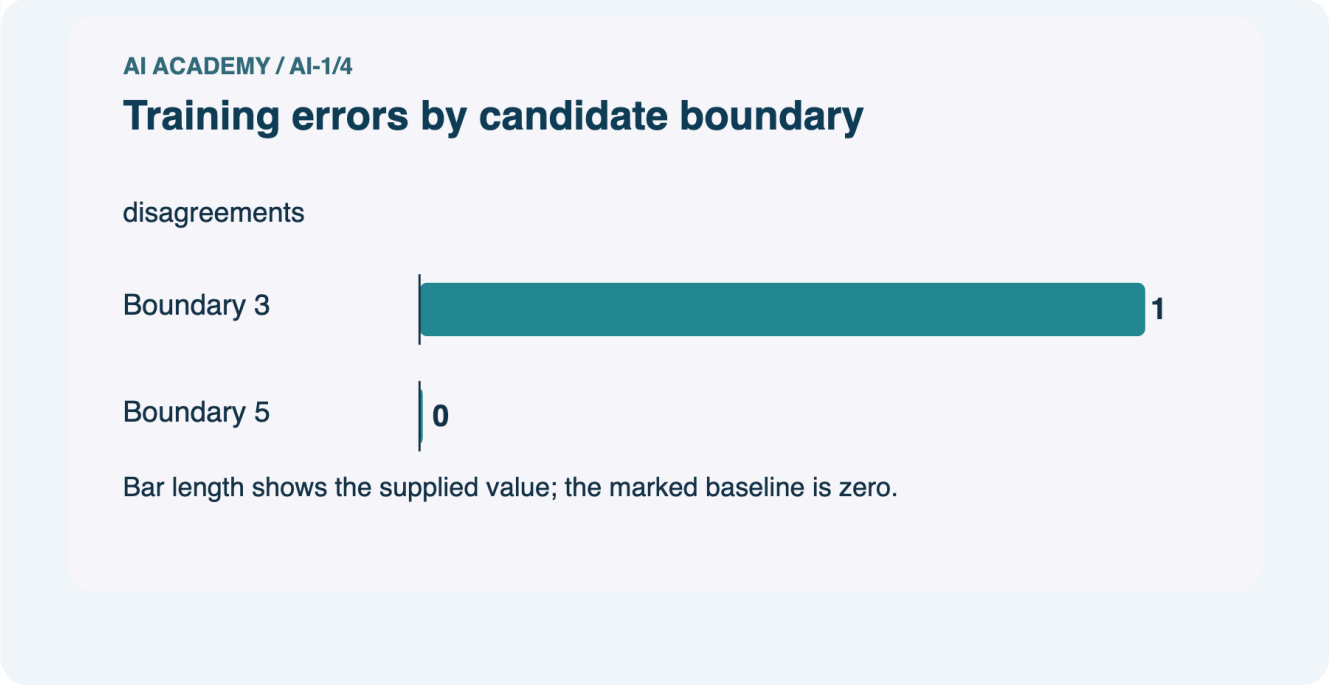
Follow the labelled picture with the matching explanation. Open the vector diagram

Training examples: Diameters 1,3,5,7 have reference **label**s S,S,L,L. → Candidate 3: **Prediction**s S,L,L,L: one disagreement at 3. → Candidate 5: **Prediction**s S,S,L,L: zero disagreements. → New pebble: Selected boundary 5 predicts small for a new diameter 4.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Training errors by candidate boundary



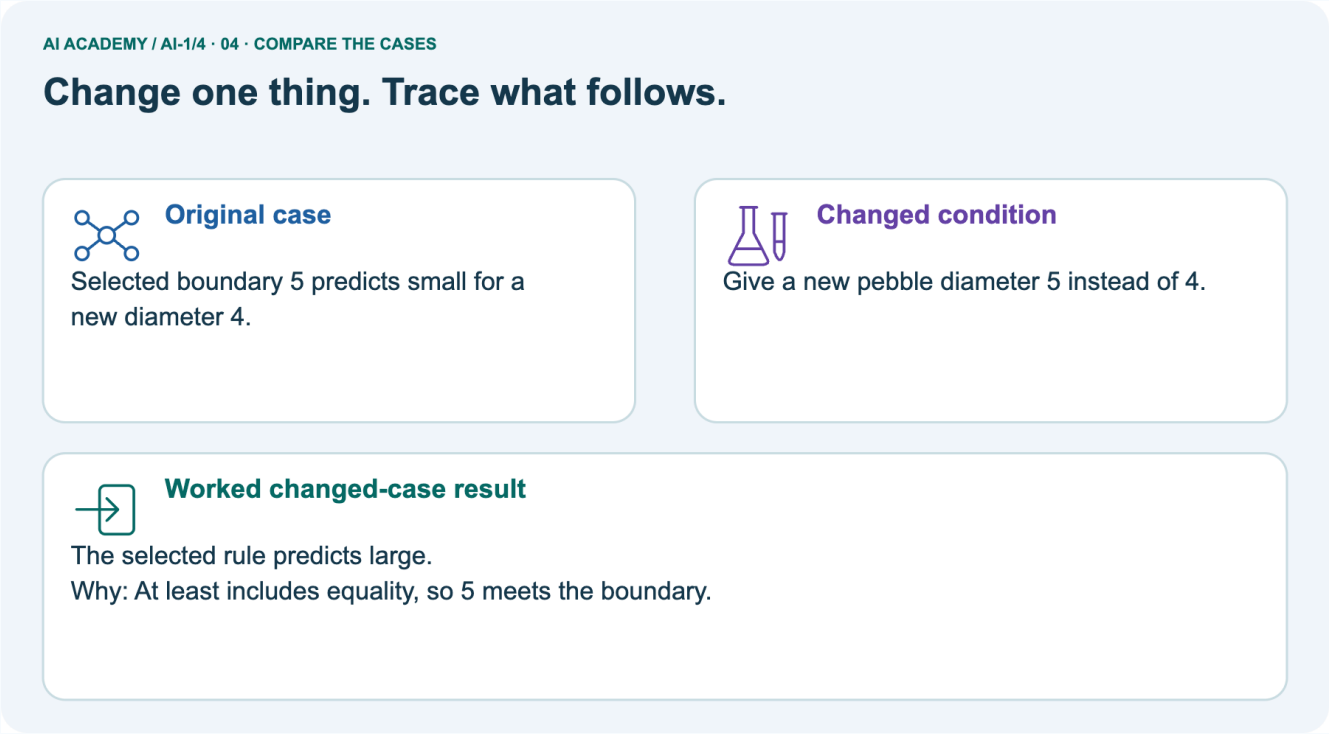
Follow the labelled picture with the matching explanation. Open the vector diagram

Training errors by candidate boundary. Units: disagreements. Boundary 3: 1. Boundary 5: 0.

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

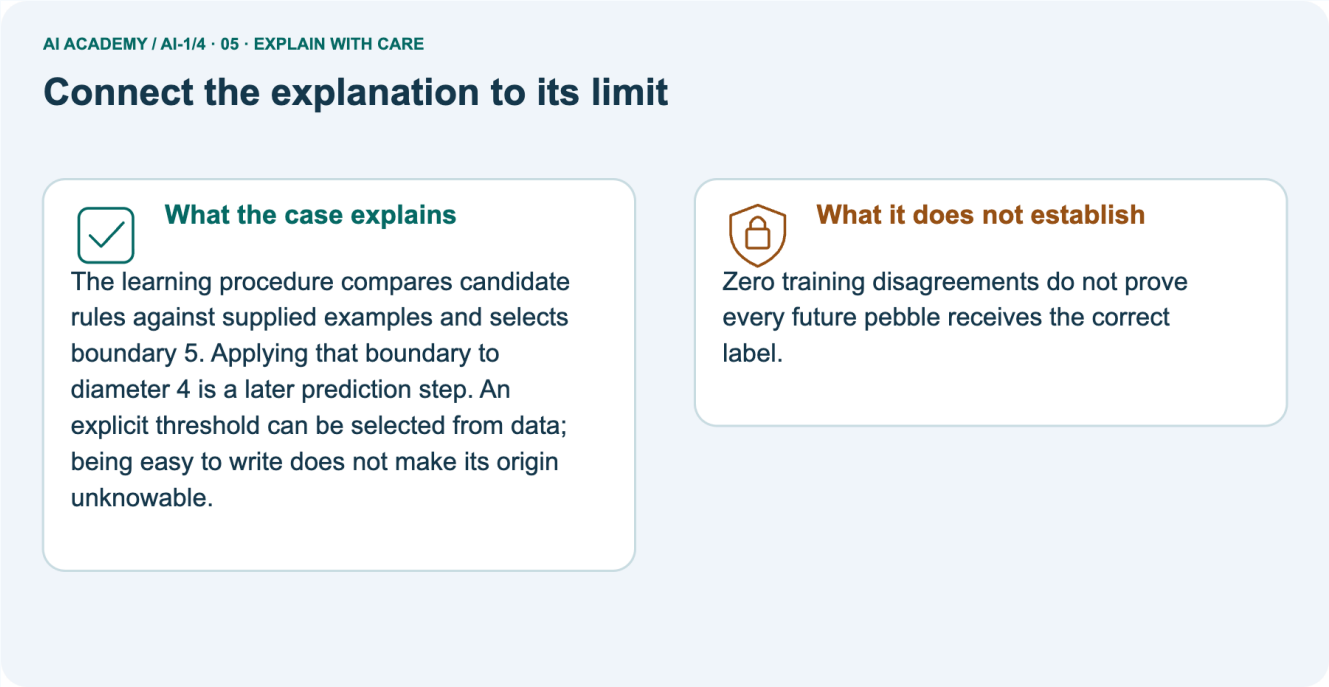
Original case: Selected boundary 5 predicts small for a new diameter 4. Changed condition: Give a new pebble diameter 5 instead of 4. Expected result: The selected rule predicts large.

Reason: At least includes equality, so 5 meets the boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

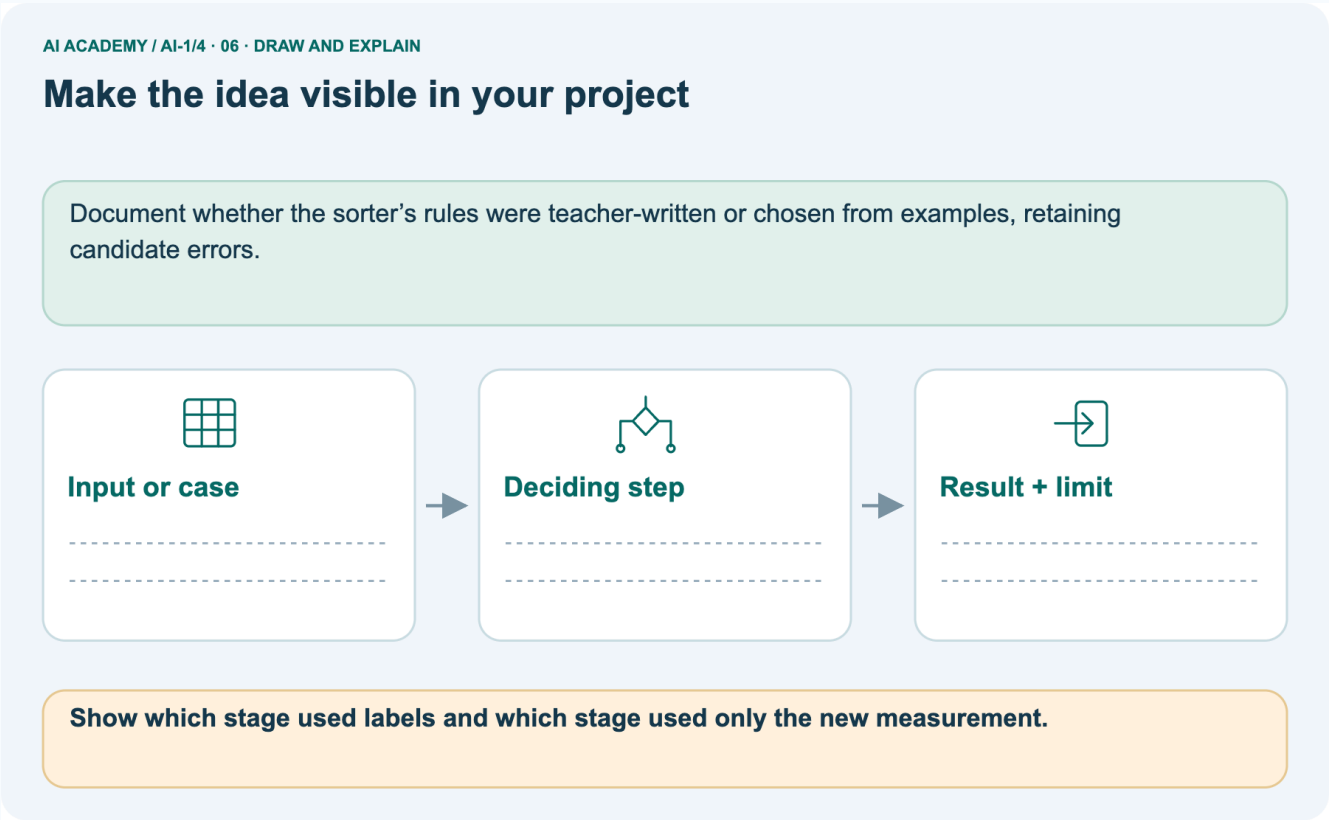
The learning procedure compares candidate rules against supplied examples and selects boundary 5. Applying that boundary to diameter 4 is a later **prediction** step.

An explicit **threshold** can be selected from data; being easy to write does not make its origin unknowable. Limit: Zero **training** disagreements do not prove every future pebble receives the correct **label**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whether the sorter's rules were teacher-written or chosen from examples, retaining candidate errors. Draw your **input**, deciding step, result and limit.

Supported question: Show which stage used **label**s and which stage used only the new measurement.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternative practice and picture transcripts

01. Rules or Learned Patterns?: the case

02. Follow the deciding steps

03. Training errors by candidate boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.